

chapter-01

July 7, 2026

1 Infinite Series, Power Series

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import math

# from IPython.display import HTML
# HTML('<style>img[alt="skan"] { filter: invert(100%); }</style>')

plt.style.use('dark_background')
```

1.1 Problems, Section 2

In the following problems, find the limit of the given sequence as $n \rightarrow \infty$.

1.1.1 2.6

$$\lim_{n \rightarrow \infty} \frac{n^n}{n!}$$

```
[2]: import matplotlib.pyplot as plt
import math

# 1. Simulation setup
# We keep the range small (1 to 20) because n^n grows insanely fast.
max_n = 20
n_values = list(range(1, max_n + 1))
results = []

# 2. Main calculation loop
for n in n_values:
    # Calculate the numerator (n to the power of n) and denominator (n!)
    ↪factorial)
    numerator = n ** n
    denominator = math.factorial(n)

    # Python handles massive integers natively, so we can just divide
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    value = numerator / denominator
    results.append(value)

# Print a few values to the console for a quick check
print("Sequence values for the first few steps:")
for i in [1, 2, 5, 10, 19]:
    print(f"For n={i}: the value is approximately {results[i-1]:.2f}")

# 3. Plotting the chart
plt.figure(figsize=(10, 6))

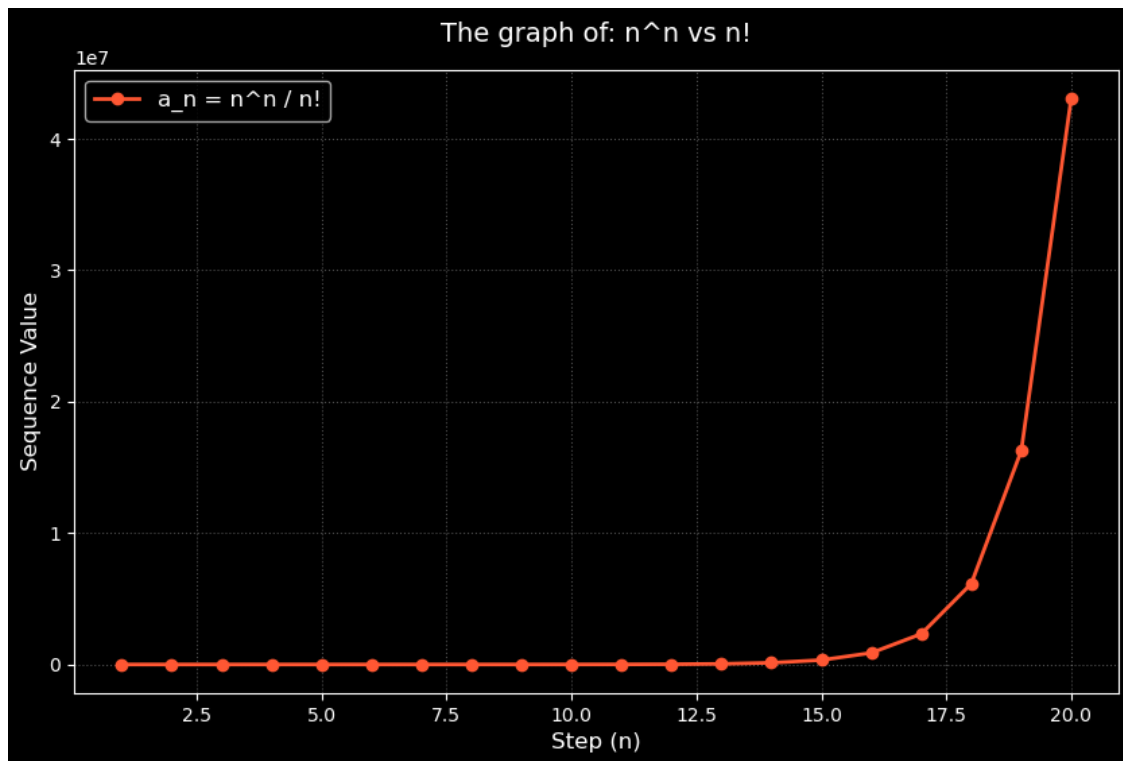
# Draw the main line
plt.plot(n_values, results, marker='o', markersize=6, color='#FF5733',
        linewidth=2, label='a_n = n^n / n!')

# Aesthetics and labels
plt.title('The graph of: n^n vs n!', fontsize=14, pad=15)
plt.xlabel('Step (n)', fontsize=12)
plt.ylabel('Sequence Value', fontsize=12)
plt.grid(True, linestyle=':', alpha=0.3)
plt.legend(fontsize=12)

# Render the plot
plt.show()

```

Sequence values for the first few steps:
 For n=1: the value is approximately 1.00
 For n=2: the value is approximately 2.00
 For n=5: the value is approximately 26.04
 For n=10: the value is approximately 2755.73
 For n=19: the value is approximately 16263866.35



$$\lim_{n \rightarrow \infty} \frac{n^n}{n!} = \infty$$

1.1.2 2.7

$$\lim_{n \rightarrow \infty} \left((1 + n^2)^{\frac{1}{\ln n}} \right)$$

```
[3]: def compute():
    max_n = 1000
    n_values = list(range(2, max_n))
    results = []

    for n in n_values:
        base = (1 + n**2)
```

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        power = 1/(math.log(n))
        value = base ** power
        results.append(value)

    for i in [2,4,8,16,32,64,128,256,512]:
        print(f"For {i} the value is approximately {results[i-1]:.4f}")

compute()

```

For 2 the value is approximately 8.1328
 For 4 the value is approximately 7.5713
 For 8 the value is approximately 7.4304
 For 16 the value is approximately 7.3981
 For 32 the value is approximately 7.3910
 For 64 the value is approximately 7.3895
 For 128 the value is approximately 7.3891
 For 256 the value is approximately 7.3891
 For 512 the value is approximately 7.3891

Handwritten derivation of the limit:

$$\lim_{n \rightarrow \infty} (1+n^2)^{\frac{1}{n^2}} = e^2$$
 The derivation uses the identity $\ln(a^b) = b \ln a$ and the expansion $\ln(1+x) \approx x$ for small x .

$$\ln((1+n^2)^{\frac{1}{n^2}}) = \frac{1}{n^2} \ln(1+n^2) = \frac{\ln(1+n^2)}{n^2}$$
 As $n \rightarrow \infty$, $\frac{\ln(1+n^2)}{n^2} \rightarrow 0$. Using the expansion $\ln(1+x) \approx x$ for small x , we get:

$$\frac{\ln(1+n^2)}{n^2} \approx \frac{n^2}{n^2} = 1$$
 Therefore,

$$\lim_{n \rightarrow \infty} (1+n^2)^{\frac{1}{n^2}} = e^1 = e$$

1.1.3 2.8

$$\lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!}$$

```

[4]: def compute():
    max_n = 1024
    n_values = list(range(2, max_n+1))
    results = []

    for n in n_values:
        numerator = math.factorial(n)**2
        denominator = math.factorial(2*n)
        value = numerator / denominator
        results.append(value)

    for i in [2,4,8,16,32,64,128,256,512, 1023]:

```

```
print(f"For {i} the value is approximately {results[i-1]:.4f}")
```

```
compute()
```

For 2 the value is approximately 0.0500
 For 4 the value is approximately 0.0040
 For 8 the value is approximately 0.0000
 For 16 the value is approximately 0.0000
 For 32 the value is approximately 0.0000
 For 64 the value is approximately 0.0000
 For 128 the value is approximately 0.0000
 For 256 the value is approximately 0.0000
 For 512 the value is approximately 0.0000
 For 1023 the value is approximately 0.0000

2.8

$$\lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!}$$

$$(n!)^2 = (1 \cdot 2 \cdot 3 \cdot \dots \cdot n)(1 \cdot 2 \cdot 3 \cdot \dots \cdot n)$$

$$(2n)! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n(n+1) \cdot (n+2) \cdot \dots \cdot 2n$$

$$a_n = \frac{n! \cdot n!}{n! \cdot (n+1)(n+2) \cdot \dots \cdot 2n} = \frac{n!}{(n+1)(n+2) \cdot \dots \cdot 2n}$$

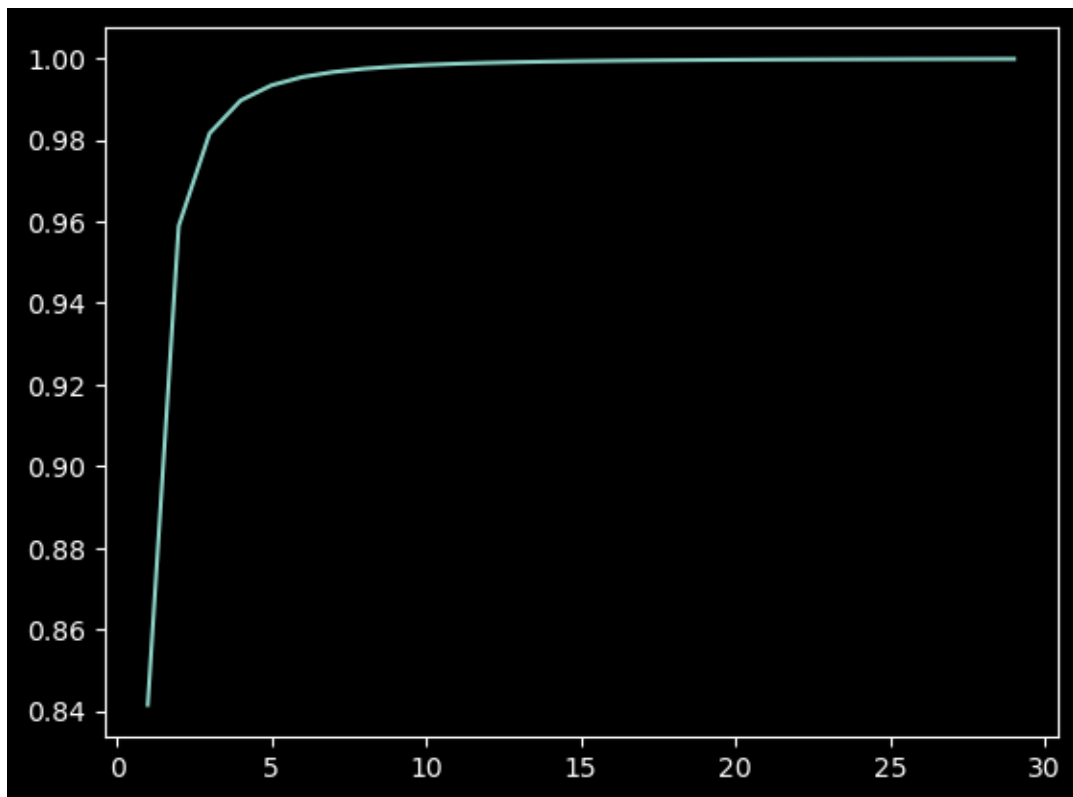
$$= \frac{1}{n+1} \cdot \frac{2}{n+2} \cdot \frac{3}{n+3} \cdot \dots \cdot \frac{n}{n+n}$$

$$a_n \leq \left(\frac{1}{2}\right)^n \Rightarrow \lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!} = 0$$

1.1.4 2.9

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right)$$

```
[5]: def compute():  
    n_max = 30  
    n_values = list(range(1,n_max))  
    results = []  
  
    for n in n_values:  
        value = n * math.sin(1/n)  
        results.append(value)  
  
    plt.plot(n_values, results)  
    plt.show()  
compute()
```



2.9

$$\lim_{n \rightarrow \infty} n \sin \frac{1}{n} = [0 \cdot \infty]$$

$\nearrow \infty$
 $\searrow 0$

$$x = \frac{1}{n} \Rightarrow n = \frac{1}{x}$$

As $n \rightarrow \infty$, $\frac{1}{x}$ approaches 0

$$\lim_{n \rightarrow \infty} n \sin \frac{1}{n} = \lim_{x \rightarrow 0} \frac{1}{x} \sin x = \lim_{x \rightarrow 0} \frac{\sin x}{x} =$$

$$= \lim_{n \rightarrow \infty} 1 \cdot \frac{\sin x}{x} = 1$$